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Rotations and translations

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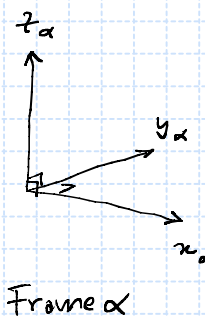
Office hours

Thursdays: 15h30-17h00

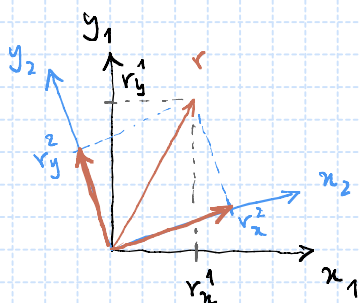
To describe vectors we need to define its components in a frame.

Orthonormal right-handed frames

- orthogonal axes
- right-handed means that $x \times y = z$
- $\|x\| = \|y\| = \|z\| = 1$



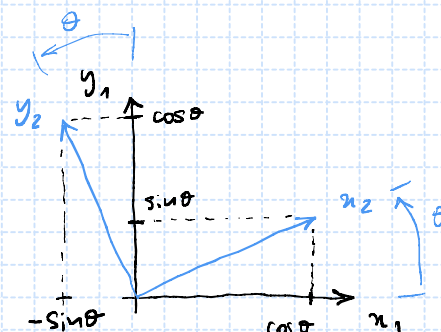
(a)



$$r^1 = \begin{pmatrix} r_{x_1}^1 \\ r_{y_1}^1 \end{pmatrix} \neq r^2 = \begin{pmatrix} r_{x_2}^2 \\ r_{y_2}^2 \end{pmatrix}$$

$$r = r_{x_2}^2 x_2 + r_{y_2}^2 y_2$$

$$x_2^1 = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad y_2^1 = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$



(b) $r^2 = \begin{pmatrix} r_{x_2}^2 \\ r_{y_2}^2 \end{pmatrix}, \quad r^1 = ?$

$$r^\alpha = r_x^\beta x_\beta^\alpha + r_y^\beta y_\beta^\alpha + r_z^\beta z_\beta^\alpha$$

$$r^\alpha = r_x^2 x_2^\alpha + r_y^2 y_2^\alpha, \quad r^1 = r_x^2 x_2^1 + r_y^2 y_2^1 = r_x^2 \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} + r_y^2 \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

(c)

$$r^1 = R_2^1 r^2$$

$$r^1 = \underbrace{\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_{R_2^1} \underbrace{\begin{pmatrix} r_x^2 \\ r_y^2 \end{pmatrix}}_{r^2}$$

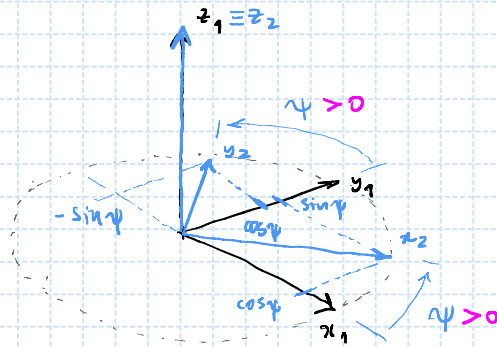
$$r^\alpha = R_\beta^\alpha r^\beta$$

change in a vector's coordinates

$$R_\beta^\alpha = \begin{pmatrix} x_\beta^\alpha & y_\beta^\alpha & z_\beta^\alpha \end{pmatrix}$$

rotation matrix

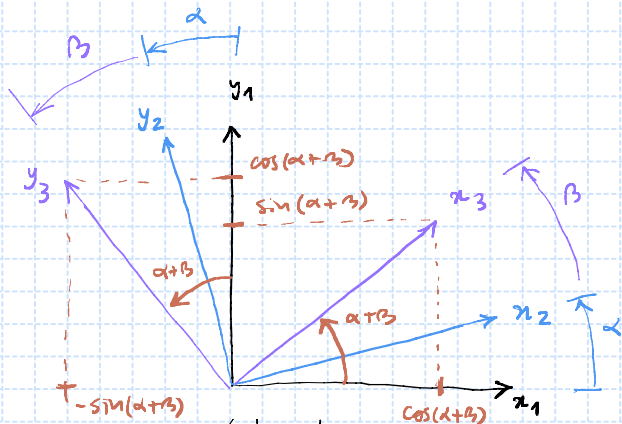
2. (a)



$$R_2^1 = \begin{pmatrix} x_2^1 & y_2^1 & z_2^1 \end{pmatrix}$$

$$R_2^1 = \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

3.



$$R_2^1 = \begin{pmatrix} x_2^1 & y_2^1 \end{pmatrix} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$R_3^2 = \begin{pmatrix} x_3^2 & y_3^2 \end{pmatrix} = \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix}$$

$$r^\alpha = R_B^\alpha r^\beta$$

$$R_2^1 R_3^2 = R_2^1 \begin{pmatrix} x_3^2 & y_3^2 \end{pmatrix} = \begin{pmatrix} R_2^1 x_3^2 & R_2^1 y_3^2 \end{pmatrix} = \begin{pmatrix} x_3^1 & y_3^1 \end{pmatrix} = R_3^1$$

$$R_B^\alpha = R_Y^\alpha R_B^\beta$$

composition of rotations

$$R_3^1 = \begin{pmatrix} x_3^1 & y_3^1 \end{pmatrix} = \begin{pmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{pmatrix}$$

$$R_2^1 = \begin{pmatrix} x_2^1 & y_2^1 \end{pmatrix} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$R_3^2 = \begin{pmatrix} x_3^2 & y_3^2 \end{pmatrix} = \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix}$$

$$R_3^1 = R_2^1 R_3^2 = \begin{pmatrix} c_\alpha & -s_\alpha \\ s_\alpha & c_\alpha \end{pmatrix} \begin{pmatrix} c_\beta & -s_\beta \\ s_\beta & c_\beta \end{pmatrix} =$$

$$= \begin{pmatrix} c_\alpha c_\beta - s_\alpha s_\beta & -c_\alpha s_\beta - s_\alpha c_\beta \\ s_\alpha c_\beta + c_\alpha s_\beta & -s_\alpha s_\beta + c_\alpha c_\beta \end{pmatrix}$$

$$\cos \alpha = c_\alpha$$

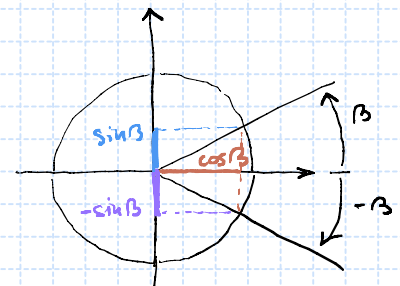
$$\sin \alpha = s_\alpha$$

$$\cos(\alpha+\beta) = c_{\alpha+\beta}$$

$$\sin(\alpha+\beta) = s_{\alpha+\beta}$$

$$\cos(\alpha-\beta) = \cos(\alpha)\cos(-\beta) - \sin(\alpha)\sin(-\beta)$$

$$\cos(\alpha-\beta) = c_\alpha c_\beta + s_\alpha s_\beta$$



$$\cos(-\beta) = \cos(\beta)$$

$$\sin(-\beta) = -\sin(\beta)$$